

<https://coda-erp-pfpm.onrender.com/>

this is what I'm currently working but I want to take a new direction:

A user can register their business first in our marketplace 

This pivot moves you from building a standard ERP to creating an **Embedded-Finance Commerce OS for MSMEs**. This is a highly lucrative direction, especially in the Nigerian market, because it solves the two biggest problems small businesses face: **lack of capital** and **friction in record-keeping**.

To make this stand out against platforms like Bumpa or Selar, the core philosophy must be **"Invisible Bookkeeping."** MSMEs won't use a platform if it feels like accounting. They *will* use it if it feels like a tool that gets them more customers and unlocks cash.

Here is a deep dive into how to architect this platform to thrive on low bandwidth, require minimal internet, and leverage AI to drive user retention.

1. Conquering the Connectivity Barrier (Low-Bandwidth & Offline First)

To win the Nigerian MSME market, your platform must work when MTN or Airtel is acting up, or when the user is out of data.

- **The WhatsApp Bot (Primary Interface):**

Instead of forcing users to download an app or open a web browser, build a WhatsApp chatbot (using APIs like Termii or Twilio).

- **To log a sale:** The user sends a message like `Sold 2 pairs of shoes for 10k`. The AI parses this and updates their inventory and ledger.
- **Receipt Uploads:** They snap a picture of a handwritten supplier receipt and send it to the bot. AI (OCR) extracts the amount, date, and items, automatically logging it as an expense.
- **Customer Storefront:** Utilize WhatsApp Business API's catalog features. Customers can browse the merchant's custom store link *inside* WhatsApp, saving them from opening a browser.

- **USSD Fallback (Zero Internet):**

For moments with absolutely no internet, integrate a USSD gateway (via Africa's Talking or similar).

- Dial `*YOUR_CODE#`
- `1. Log Cash Sale` (Updates their AI credit score instantly).

- 2. Check Inventory
- 3. Request Overdraft/Loan
- **Progressive Web App (PWA):**
Build the web dashboard as a PWA (easily hosted on Vercel). Using Service Workers, the web app caches on their phone. They can open the app, log sales, and generate invoices completely offline. The moment their phone detects an EDGE or 3G connection, the app silently syncs the data to your servers in the background.

2. The "Trojan Horse" Lending Engine (AI & Banking)

Your biggest differentiator is that the e-commerce store is actually a data-gathering engine for micro-lending.

- **Dynamic Virtual Accounts:** Partner with a payment processor (like Paystack) or a BaaS (Banking-as-a-Service) provider to issue a dedicated virtual account number to every registered business. When customers pay into this account, the transaction is automatically logged in the merchant's books.
- **Tiered KYC & Features:**
 - *Tier 0 (No BVN/NIN):* Can create a store link, add products, and log cash sales manually. No banking features.
 - *Tier 1 (BVN Provided):* Unlocks the virtual account to receive transfers. The AI begins tracking cash flow.
 - *Tier 2 (Full NIN/CAC if applicable):* Unlocks the lending dashboard.
- **AI "Micro-Testing" Algorithms:**
Since new users start with a credit score of 0, traditional banks won't touch them. Your AI evaluates alternative data: *How consistently do they log sales? How fast does their inventory turn over? Do they have repeat customers?*
 - **The Trust Ladder:** The system auto-offers a tiny, low-risk overdraft (e.g., ₦5,000 for 3 days). If they pay it back via their storefront revenue, their internal score goes up, and the next offer is ₦15,000.

3. Gamification: Rewarding the "Lazy" Business Owner

Small business owners don't want to track books because the reward is usually abstract (a spreadsheet). You must make the reward immediate and financial.

- **The "Capital Readiness" Progress Bar:**
When a user logs in, the first thing they should see is a large, colorful gauge. Every time they upload a receipt, create an invoice, or log a sale via WhatsApp, the gauge fills up.
 - *Message:* "You are 70% of the way to unlocking a ₦50,000 overdraft! Log 3 more expenses to reach 100%."

- **Instant Gratification Prompts:**
When they upload a picture of a receipt, don't just say "Saved." Say, "Receipt saved! Your credit health score just increased by 2 points." This creates a dopamine loop similar to social media, but for business management.
- **One-Click Invoicing:**
Remove all technical jargon. A button that says "Get Paid Now" lets them type an amount, and it instantly generates a lightweight PDF or a WhatsApp-ready link that they can forward to a client.

4. Technical Architecture Strategy

To keep the infrastructure lightweight and fast:

- **Frontend Stores (Customer-facing):** Build these as highly optimized, statically generated pages. A customer clicking a merchant's link should load the page in under 2 seconds on a 3G connection. Strip out heavy JavaScript; rely on pure HTML/CSS where possible.
- **Backend & AI:**
 - Use a lightweight Node.js or Python backend.
 - For the AI receipt scanning, you can use Google Cloud Vision API or OpenAI's multimodal models to extract text from unstructured images (receipts, handwritten notes).
 - For the credit scoring, start with a simple rules-based algorithm (weights assigned to frequency of sales, volume, and repayment speed) before training complex machine learning models.

By stripping away the concept of an "ERP" and positioning this as a "**Smart Store that pays you to use it**," you drastically lower the barrier to entry. The combination of WhatsApp integration, USSD fallbacks, and instant micro-lending creates a sticky ecosystem that will be very difficult for competitors to replicate.

To explore the technical execution:

How to set up the WhatsApp bot flow



Design the AI credit scoring logic



Implementing Tiered KYC in Nigeria



turn this into a detailed prd and roadmap leading with the marketplace

Here is the Product Requirements Document (PRD) and technical roadmap for your Embedded-Finance Commerce OS. It is designed around the 2026 Nigerian regulatory landscape and optimized for a lightweight, progressive tech stack.

PRD: Embedded-Finance Commerce OS for MSMEs

1. Product Vision & Strategy

The Problem: Small businesses in emerging markets struggle with two interconnected issues: friction in logging daily transactions and total exclusion from formal credit due to a lack of financial data.

The Solution: A low-bandwidth, offline-capable storefront that acts as a "Trojan Horse." By providing an incredibly easy way to sell online and via WhatsApp, the platform invisibly builds a verifiable ledger. This ledger powers a gamified credit-scoring engine, unlocking micro-loans and overdrafts.

Core Philosophy: "Invisible Bookkeeping." If the merchant feels like they are doing accounting, the product has failed. Every action must feel like either *getting a customer* or *getting capital*.

2. Target Audience

- **Primary User:** Micro and Small Enterprise (MSME) owners operating heavily on mobile.
- **Key Traits:** Low tolerance for technical jargon, experiences frequent internet fluctuations, prioritizes immediate cash flow over long-term analytics.

3. Core Features & Specifications

3.1. The Zero-Friction Marketplace (Phase 1)

The foundational hook. Merchants must be able to set up a store in under 60 seconds.

- **PWA Merchant Dashboard:** A Progressive Web App hosted on edge infrastructure (like Vercel) that aggressively caches offline. Merchants can add products, set prices, and view their catalog without an active internet connection. Syncs automatically upon reconnection.

- **WhatsApp Storefront Bot:** Customers can view catalogs and place orders directly inside WhatsApp. Merchants can log cash sales by texting the bot (e.g., `Sold 2 bags of rice for 20k`).
- **USSD Fallback:** A dedicated `*CODE#` for feature-phone users or zero-data environments to log cash sales and check their capital readiness score.

3.2. Automated Bookkeeping & Receipt AI (Phase 2)

Removing the manual data entry bottleneck.

- **One-Click Invoicing:** Generate lightweight PDF invoices or direct payment links to share via WhatsApp.
- **AI Expense Logging:** Merchants upload photos of handwritten receipts or supplier invoices. The system uses Optical Character Recognition (OCR) to parse the date, amount, and items, auto-updating the expense ledger.

3.3. Embedded Banking & CBN 2026 KYC Compliance (Phase 3)

The financial engine. To comply with the 2026 Central Bank of Nigeria (CBN) mandates, onboarding is strictly tiered.

KYC Level	Requirement	Daily Limit	Unlocked Features
Tier 0	Phone Number / WhatsApp	₦0 (No Banking)	Storefront catalog, manual cash sales, inventory management
Tier 1	Verified BVN or NIN	₦30,000	Dedicated virtual account (via Paystack/Bank), automated payments, auto-expense ledger.
Tier 2	Verified Govt ID + Address	₦500,000	Access to "Credit Readiness" score, micro-overdraft, marketplace integration

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- **Dynamic Virtual Accounts:** Upon Tier 1 verification, the merchant is assigned a unique virtual account number. Inbound transfers automatically reconcile against pending

invoices or register as generic storefront sales.

- **Gamified Capital Readiness:** A prominent visual gauge tracking their internal credit score. Actions like logging 5 consecutive days of sales, receiving a virtual account transfer, or verifying a NIN increase the score and move them closer to loan eligibility.

4. Technical Architecture Strategy

To maintain high performance and low operational costs:

- **Frontend (Merchant & Customer):** Next.js Progressive Web App deployed on Vercel. Static generation for storefront pages to ensure sub-2-second load times on 3G networks.
- **Backend & APIs:** Node.js/Express or Python/FastAPI microservices.
 - *Identity Service:* Integration with platforms like Interswitch or EaseVerifier for real-time BVN/NIN validation.
 - *Payments Service:* Paystack API for virtual account creation and webhook listening for auto-reconciliation.
 - *Messaging Service:* Twilio or Termii API for the WhatsApp bot and USSD gateway routing.
- **Database:** PostgreSQL for robust transactional integrity and JSONB support for flexible product catalogs.

5. Execution Roadmap

○ Phase 1: The Marketplace & Offline Foundation

Months 1 – 2

Focus: Getting merchants to use the platform for operations before introducing finance.

- Build the Next.js PWA with offline caching (Service Workers).
- Develop the core database schema (Users, Products, Transactions).
- Launch the WhatsApp Bot for catalog viewing and manual sale logging.
- Implement basic USSD routing for offline cash logging.

○ Phase 2: Embedded Payments & AI Ledger

Months 3 – 4

Focus: Automating the books and capturing digital cash flow.

- Integrate Tier 1 KYC (BVN/NIN real-time lookup APIs).

- Integrate Paystack virtual accounts linked to the merchant's profile.
- Develop webhook listeners to auto-update the ledger when a transfer hits the virtual account.
- Roll out the AI receipt scanner (OCR integration for expense tracking).

○ Phase 3: The Lending Engine & Gamification

Months 5 – 6

Focus: Activating the "Trojan Horse" credit system to drive retention.

- Implement Tier 2 KYC upgrades (ID and address verification).
- Build the "Capital Readiness" scoring algorithm based on transaction velocity and ledger consistency.
- Launch the visual progress bar dashboard for merchants.
- Roll out the first test cohort for micro-overdrafts (₦5,000 - ₦15,000) funded internally or via a lending partner.

Which technical component should we define next?

Design the Paystack Virtual Account flow



Structure the WhatsApp Bot architecture



Draft the Database Schema

